WITNESS-4 Preregistration

“The Freshness Bracket” — Non-Backdatable, Independently-Reconstructable, Chain-Linked Quantum ProofRecord

Series: WITNESS (fourth record — append-only ledger W1 → W2 → W3 → W4)
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0. One-paragraph summary

WITNESS-4 asks a narrow, high-value question that an auditor or independent reviewer would actually raise about any ProofRecord™: “Could this record have been backdated or pre-computed, and can I rebuild it myself from public data alone?” WITNESS-4 answers both by constructing a **freshness bracket** around the record. First, the entire experimental design (CHSH circuit spec, declared intent, context id, and the previous ledger link) is written into a **pre-commitment document and hashed BEFORE any unpredictable public anchor is fetched** — establishing design-before-anchor. Second, the fused nonce commits to a **NIST public Randomness Beacon** pulse whose value did not exist before its published pulse time — establishing a not-before lower time bound (the record could not have been finalized earlier). Third, the record is **chain-linked to WITNESS-3’s published record_hash** (Zenodo DOI 10.5281/zenodo.21434832), making WITNESS-1..4 a single append-only, third-party-verifiable ledger. Fourth, a dedicated reconstruction module rebuilds the entire record — every hash, the fused nonce, the CHSH statistic, the freshness bracket, and the record_hash — from the stored public artifacts alone, with zero trust in any RF-private state. The same QPU run remains a CHSH Bell test, certified under the identical honest standard used in WITNESS-3.

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1. Claim (single, falsifiable)

A ProofRecord constructed with a length-prefixed **7-segment fused nonce** — binding a pre-commitment hash, the previous ledger link, QPU CHSH counts, the provider job id, the backend calibration, a NIST beacon pulse, and a byte-exact LIGO/GWOSC strain segment — can be:

(a) **fully reconstructed** from the stored public artifacts alone by an independent party, with the provider job record confirming the QPU counts when the provider API is reachable (W4-C1);
(b) detected as tampered when **any single element** of any witness, the pre-commitment, or the ledger link is substituted (W4-C2);
(c) shown to be **non-backdatable and non-pre-computable** by re-evaluating the freshness

bracket independently of the record's own stored freshness block (W4-C3);
(d) shown to be a sound **append-only ledger link** to WITNESS-3, with any broken link detected three independent ways (W4-C4); and
(e) accompanied by a **CHSH Bell-inequality violation** certified at ≥ 5σ above the classical bound on the tested backend (W4-C5),

under the tested construction, circuit, backend, qubits, calibration, and shot count.

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2. Commercial motivation (why this record, for RF Inc.)

WITNESS-4 directly answers **Prospect Question #5** — "Can an independent reviewer reconstruct the decision from the ProofRecord?" — with a machine that does exactly that, offline, from public artifacts. It also answers the single most common audit/compliance objection to any evidentiary record: **"How do I know it wasn't backdated or pre-computed after the fact?"** By making the freshness bracket and the append-only ledger link part of the nonce itself, WITNESS-4 strengthens the utility case for the **ProofRecord™** construction and supports the record-integrity posture of the RF-100 program. This is a provenance / freshness / ordering advance, not a new physics claim.

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3. The freshness bracket (precise definition)

All timestamps are ISO-8601 UTC. The bracket is:

```
    precommit_time_utc   <=   nist_pulse_time   <=   timestamp_utc (finalize)
```

and the verifier checks three invariants:

1. **NOT-BEFORE bound** — `timestamp_utc >= nist_pulse_time`. The fused nonce commits to a NIST beacon `outputValue` that was not published before `nist_pulse_time`; a record claiming an effective time earlier than the pulse is a backdating forgery.
2. **DESIGN-BEFORE-ANCHOR** — `precommit_time_utc <= nist_pulse_time`. The pre-commitment (circuit + intent + context + prev link) was hashed before the beacon value was known; a `precommit_time` after the pulse would indicate the design could have been chosen with knowledge of the anchor (pre-computation).
3. **CHAIN MONOTONIC** (advisory) — `timestamp_utc >= previous ledger record's timestamp`.

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4. Experimental cases and PASS criteria

W4-C1 — Zero-trust independent reconstruction (+ provider confirmation)

Rebuild the entire record from the public artifacts (`raw_counts`, `calibration_snapshot`, `nist_witness`, `astro_witness`, `precommit`) and compare to the stored ProofRecord.
PASS: every reconstruction check matches AND `freshness.fresh == True` AND the provider API confirms the `job_id` and per-setting counts. **GATE-STOP** (not FAIL) if the provider

API is unreachable/unauthorized at verification time (the offline reconstruction still runs and is reported).

W4-C2 — Tamper detection (8 sub-trials)

Alter exactly one element and confirm detection: (a) raw_counts, (b) job_id, (c) calibration, (d) NIST pulse, (e) astro file hash, (f) context_id without record_hash recompute, (g) pre-commitment document, (h) previous ledger link.
PASS: all 8 forgeries detected.

W4-C3 — Backdating / pre-computation detection (flagship)

Independently re-evaluate the freshness bracket from the record's timestamps (not the stored freshness block). **PASS:** a backdated `timestamp_utc` (earlier than the pulse) is detected, a pre-computed `precommit_time_utc` (later than the pulse) is detected, AND the genuine record evaluates `fresh == True` (no false positive).

W4-C4 — Append-only ledger chain integrity

Verify the link to WITNESS-3's published record_hash. **PASS:** the honest link verifies, a broken link is detected three ways (chain-link check, fused-nonce mismatch, record_hash mismatch), and the timestamps are monotonic.

W4-C5 — CHSH Bell-inequality violation certification (device-dependent)

Identical criterion to WITNESS-3 W3-C4. **PASS (all must hold):** (1) $|S| > 2$; (2) $(|S| - 2)/\sigma_S \geq 5.0$; (3) $|S| \leq 2\sqrt{2} + 0.10$. **KILL condition:** $|S| \leq 2 \rightarrow$ W4-C5 = FAIL, `bell_certified = False`, published as-is; the fused nonce and freshness bracket remain valid as provenance/freshness/ordering constructions regardless.

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5. Fused nonce construction

```
fused_nonce = SHA-256(
      LP(precommit_hash)
    || LP(prev_record_hash)
    || LP(canonical_json(raw_counts))
    || LP(job_id)
    || LP(canonical_json(calibration_snapshot))
    || LP(canonical_json(nist_beacon_record))
    || LP(canonical_json(astro_witness_record))
)
```

`LP(x) = struct.pack('>I', len(x)) + x` (4-byte big-endian length prefix; boundary-collision safe). `canonical_json` = sorted keys, compact separators, ASCII. No single input controls the nonce, and the nonce is inseparable from both the pre-commitment and the previous ledger link.

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6. Preregistered non-claims (honesty bounds)

The following are stated in advance and will NOT be claimed regardless of outcome:

- **Not a trusted timestamping / notarization authority and not a blockchain.** The freshness bracket establishes a **relative ordering** — design fixed before anchor, record finalized at-or-after the anchor's published pulse time — and a **not-before lower time bound only.** There is **no upper time bound**: a record can always be finalized later.
- **Not a loophole-free or device-independent Bell test.** The two qubits are neighbouring transmons on one chip (locality/communication loophole open), outcomes use fair sampling (detection loophole open), and settings are fixed by the harness (freedom-of-choice loophole open). A violation is consistent with non-classical correlations on the tested device under no-signalling and fair-sampling assumptions; it is not a device-independent certification.
- **Astro data is provenance-only.** The LIGO/GWOSC segment binds a fixed, public, byte-exact artifact into the nonce; it contributes no physical randomness claim.
- **No min-entropy claim from any single source.** The nonce is a provenance / freshness / ordering construction, not an information-theoretic randomness extractor.
- **No legal, security-certification, production-readiness, or RF-100-certification claims.** All claims are bounded to the tested construction and conditions.
- **record_hash provides field integrity only;** cross-context authorization enforcement is delegated to the ARK-457 layer, unchanged from prior records.

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7. Method summary (load-bearing step order)

1. Build and hash the **pre-commitment** (`precommit.json`, `precommit_hash`) — BEFORE any anchor is fetched.
2. Submit the CHSH Batch job to IBM Quantum (channel `ibm_cloud`, 4 settings, qubits 0 & 1, 2000 shots/setting, transpile opt_level=1). Capture `raw_counts`, `calibration_snapshot`, `job_meta`.
3. **After** results return, fetch the NIST beacon pulse and the LIGO/GWOSC strain witness.
4. Compute the fused nonce, evaluate the freshness bracket, and build the v4 ProofRecord (schema `witness-proofrecord-4.0`), chain-linked to WITNESS-3's published record_hash.
5. Run W4-C1..C5 and record honest verdicts (PASS / FAIL / GATE-STOP).

Genesis ledger link (WITNESS-3 published record_hash):
`858ffd49fd7517fd58ef242226bd7c680bb5db5850a34256fedffd76df0f1caf`

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8. Integrity lock

This preregistration and all `witness-4/src/*.py` source files are hashed into `witness-4/MANIFEST.sha256` and committed **before** QPU execution. The QPU run is performed only after this lock and only with the author's explicit approval.

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